

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

Project: **AI-POWERED CAREER INTELLIGENCE PLATFORM**

Milestone 1: Project Foundation and Resume Management System

SUBMITTED BY:

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Date: 12 July 2026

1. INTRODUCTION

The **AI Career Intelligence Platform** is a full-stack web application designed to help students and job seekers improve their resumes and prepare for better career opportunities.

The platform allows users to register securely, upload resumes in PDF format, analyze resume content, identify technical skills, detect missing skills, calculate a resume score, recommend suitable career paths, estimate salary ranges, and maintain a history of resume analyses.

The application combines Artificial Intelligence concepts with modern web development technologies to provide an intelligent and user-friendly career guidance system.

2. OBJECTIVE

The primary objectives of the project are:

- Develop a secure user authentication system.
- Allow users to upload resumes.
- Extract important information from resumes.
- Detect technical skills automatically.
- Calculate a resume quality score.
- Identify missing industry-relevant skills.
- Recommend suitable career paths.
- Predict expected salary ranges.
- Store resume analysis history.
- Provide a clean and user-friendly interface.

3. TECH STACK

The development of Milestone 1 utilized the following technologies and frameworks:

Frontend

- React.js
- Tailwind CSS
- Axios
- React Router DOM

Backend

- FastAPI
- Python

Database

- PostgreSQL

ORM

- SQLAlchemy

Authentication

- JWT (JSON Web Token)

API Testing

- Swagger UI

Version Control

- Git & GitHub

4. CODE IMPLEMENTATION

4.1 Frontend Implementation (React.js)

The frontend of the AI Career Intelligence Platform is developed using **React.js** and **Tailwind CSS**. React components are used to create a responsive and interactive user interface, while Tailwind CSS provides a modern and consistent design.

The frontend communicates with the FastAPI backend through REST APIs using the Axios library.

Major Components

- Home Page
- Register Page
- Login Page
- Dashboard
- Resume Analysis
- Resume History
- User Profile

Sample Code

```
const response = await axios.post(
  "http://127.0.0.1:8000/auth/login",
  {
    email,
    password
  }
);
```

```
localStorage.setItem("token", response.data.access_token);
```

```
navigate("/dashboard");
```

The above code sends user login credentials to the FastAPI backend and stores the JWT token after successful authentication.

4.2 Backend Implementation (FastAPI)

The backend is implemented using **FastAPI**, which provides high-performance REST APIs for user authentication and resume analysis.

The backend handles:

- User Registration
- User Login
- JWT Authentication
- Resume Upload
- Resume Parsing
- Resume Analysis
- Database Operations

Registration API

```
@router.post("/register")
```

```
def register(user: UserRegister, db: Session = Depends(get_db)):
```

```
    ...
```

This API creates a new user account after hashing the user's password and stores the user details in PostgreSQL.

Login API

```
@router.post("/login")
```

```
def login(user: UserLogin, db: Session = Depends(get_db)):
```

```
    ...
```

This API verifies user credentials and generates a JWT Access Token for secure authentication.

4.3 Database Implementation (PostgreSQL)

The application uses **PostgreSQL** as the relational database, and **SQLAlchemy ORM** is used for database interactions.

Users Table

```
class User(Base):
```

```
    __tablename__ = "users"
```

```
    id = Column(Integer, primary_key=True)
```

```
    full_name = Column(String)
```

```
email = Column(String)
password = Column(String)
college = Column(String)
branch = Column(String)
```

The Users table stores the registered user information securely.

Resume Table

```
class Resume(Base):
    __tablename__ = "resumes"

    id = Column(Integer, primary_key=True)
    filename = Column(String)
    resume_score = Column(Integer)
    career = Column(String)
    salary = Column(String)
```

The Resume table stores uploaded resume details along with AI-generated analysis results.

4.4 Resume Analysis Implementation

The Resume Analysis module is the core component of the application.

It performs the following tasks:

- Upload Resume
- Extract Resume Text
- Extract Email
- Extract Phone Number
- Detect Skills
- Calculate Resume Score
- Detect Missing Skills
- Recommend Career
- Predict Salary
- Store Results in PostgreSQL

Sample Code

```
text = extract_text(filepath)
```

```
email = extract_email(text)
```

```
phone = extract_phone(text)
```

```
skills = extract_skills(text)
```

```
resume_score = calculate_resume_score(skills, text)
```

This module extracts useful information from the uploaded resume and generates intelligent insights.

4.5 Authentication Implementation

JWT (JSON Web Token) authentication is implemented to provide secure login and access control.

Token Generation

```
access_token = create_access_token(  
    data={"sub": db_user.email}  
)
```

The generated JWT token is returned to the frontend after successful login.

4.6 API Integration

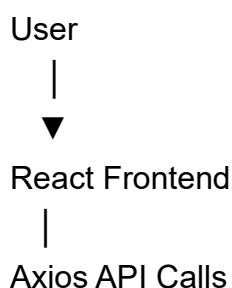
React.js communicates with the FastAPI backend using Axios.

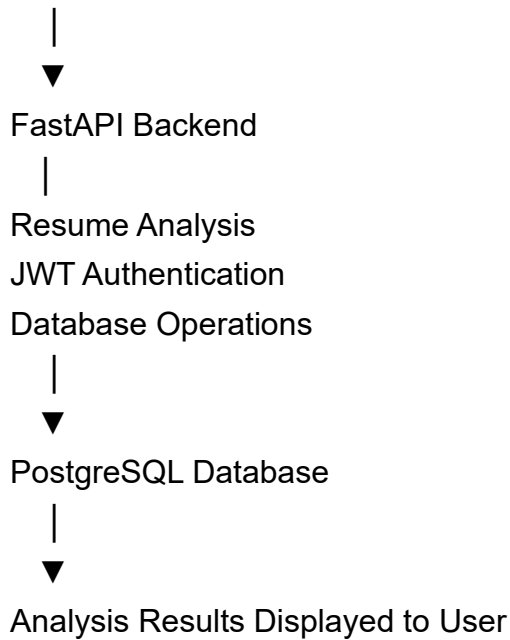
Example

```
const response = await axios.post(  
    "http://127.0.0.1:8000/resume/upload",  
    formData,  
    {  
        headers: {  
            "Content-Type": "multipart/form-data"  
        }  
    }  
);
```

The backend processes the uploaded resume and returns the complete analysis report to the frontend.

4.7 Overall Workflow





5. Screenshots and Output

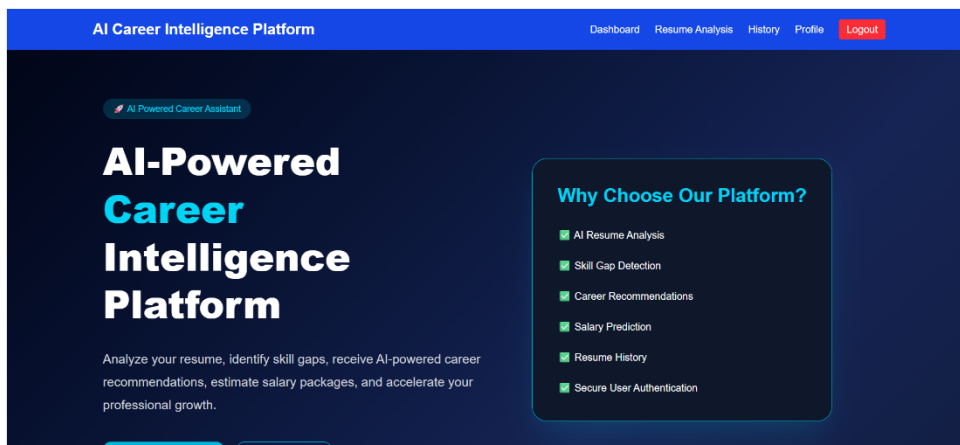
This section presents the graphical user interface (GUI) of the AI Career Intelligence Platform along with the output generated during execution. The screenshots demonstrate the successful implementation of various modules developed in Milestone 1.

5.1 Home Page

Description:

The Home Page is the landing page of the application. It provides an overview of the AI Career Intelligence Platform, highlights its key features, and allows users to navigate to different modules such as Dashboard, Resume Analysis, History, Profile, and Logout.

Screenshot:

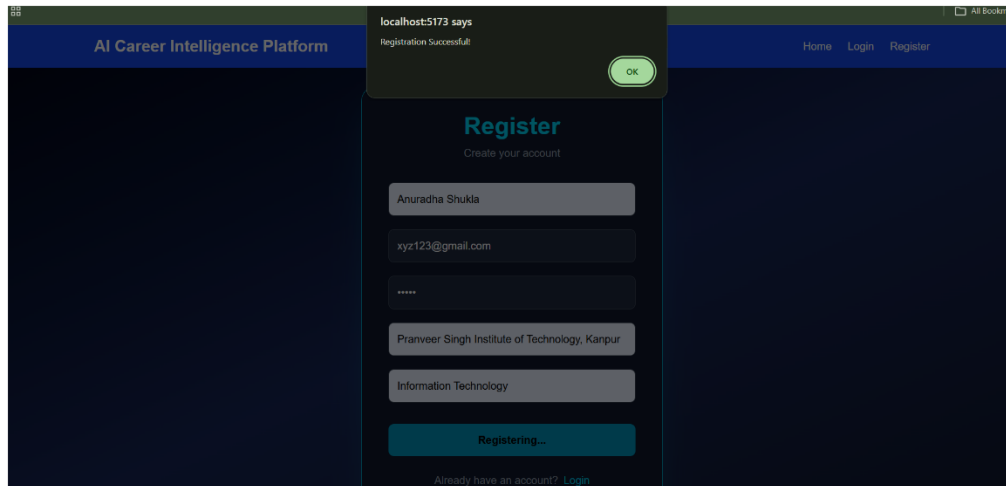


5.2 User Registration

Description:

The Registration page allows new users to create an account by entering their full name, email address, password, college, and branch. User details are securely stored in the PostgreSQL database after password hashing.

Screenshot:

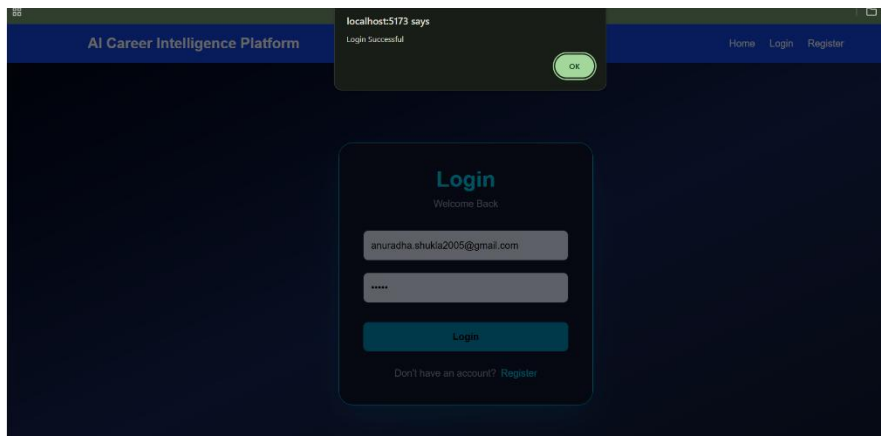


5.3 User Login

Description:

The Login page authenticates users using JWT Authentication. After successful login, users are redirected to the Dashboard where they can access all platform features.

Screenshot:



5.4 Dashboard

Description:

The Dashboard serves as the central navigation page of the application. It provides quick access to Resume Analysis, Resume History, User Profile, and Logout.

Screenshot:



5.5 Resume Analysis

Description:

The Resume Analysis page allows users to upload a PDF resume. The backend extracts text, identifies technical skills, calculates the resume score, detects missing skills, predicts salary, and recommends suitable career paths.

Screenshot:



5.6 Resume Analysis Output

Description:

After successful analysis, the application displays the extracted email, phone number, identified skills, missing skills, resume score, career recommendation, salary prediction, and resume preview.

Screenshot:

Upload Resume

Choose File 2301640130028_Anuradha Shukla (2).pdf

2301640130028_Anuradha Shukla (2).pdf

Analyze Resume

Analysis Result

Resume Score

100

Career

AI / Machine Learning Engineer

Salary

12 - 18 LPA

Email

anuradha.shukla2005@gmail.com

Phone

+91-9653030714

Skills

c

css

git

html

java

javascript

machine learning

mongodb

node

python

react

Missing Skills

sql

docker

aws

fastapi

Resume Preview

Anuradha Shukla
* Kanpur, Uttar Pradesh (cid:131) +91-9653030714 # anuradha.shukla2005@gmail.com (cid:239)
linkedin.com/in/anuradhashukla27 § github.com/Anuradha-Shukla LeetCode
OBJECTIVE
Aspiring Software QA Engineer with strong programming skills in Python and Java and a solid foundation in Software Testing, Data Structures, and Object-Oriented Programming. Passionate about building reliable software through testing, debugging, and analytical problem-solving.
EDUCATION
Pranveer Singh Institute

5.7 Resume History

Description:

The Resume History page displays all previously analyzed resumes along with their corresponding resume scores, detected skills, and career recommendations stored in the PostgreSQL database.

Screenshot:

Resume History

2301640130028_Anuradha_Shukla (47).pdf

Score: 100

Career: AI / Machine Learning Engineer

Salary: 12 - 18 LPA

Skills

c

css

deep learning

git

html

java

javascript

machine learning

mongodb

node

python

react

Missing Skills

sql

docker

aws

fastapi

2301640130028_Anuradha_Shukla (53).pdf

Score: 100

Career: AI / Machine Learning Engineer

Salary: 12 - 18 LPA

Skills

c

css

git

html

java

javascript

machine learning

mongodb

node

python

react

Missing Skills

sql

docker

aws

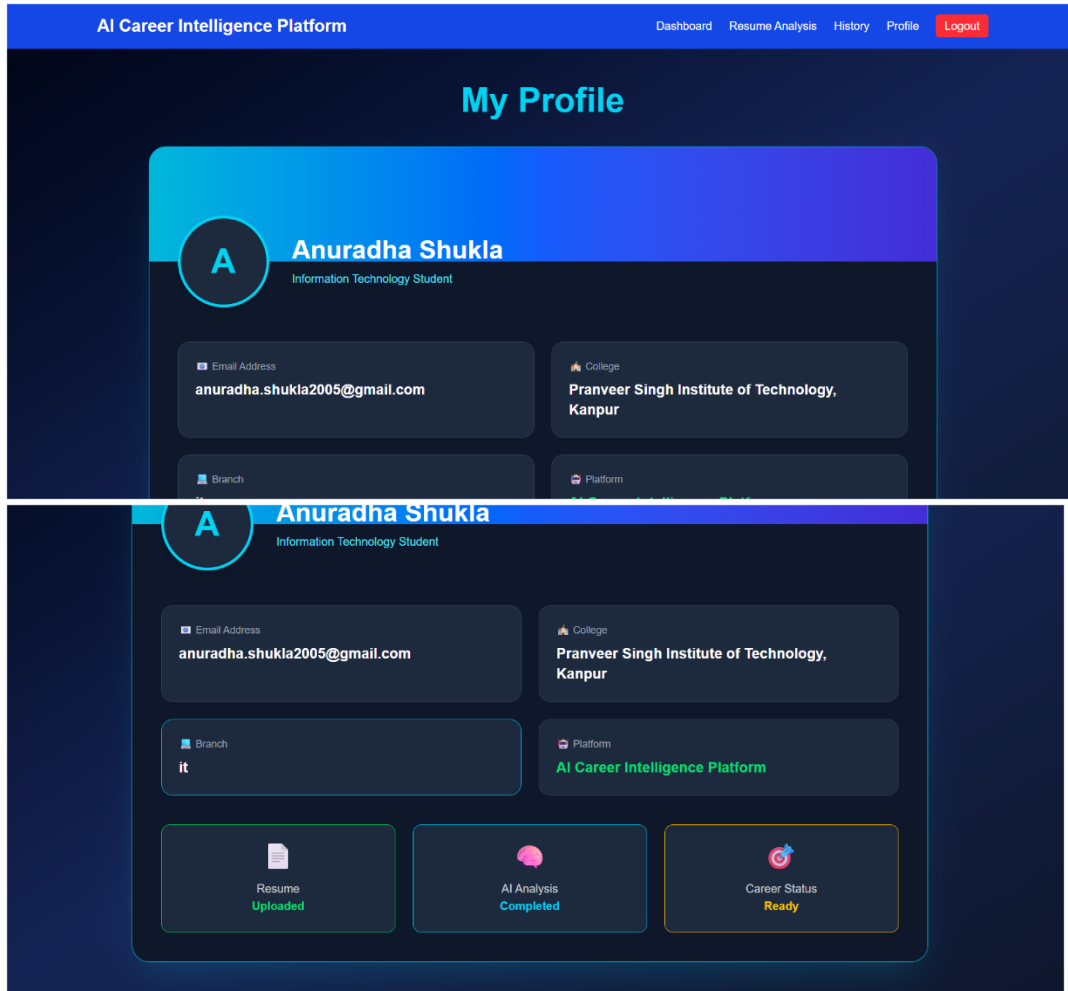
fastapi

5.8 User Profile

Description:

The Profile page displays the authenticated user's personal information, including full name, email address, college, and branch.

Screenshot:

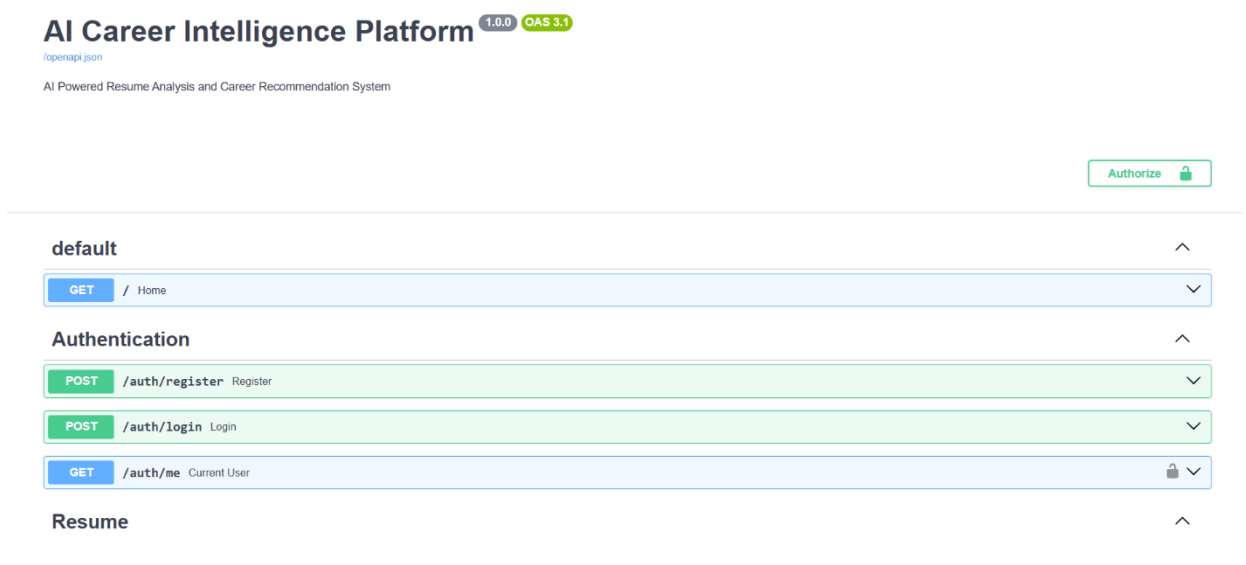


5.9 Swagger API Documentation

Description:

FastAPI automatically generates interactive API documentation using Swagger UI. It allows developers to test API endpoints such as registration, login, resume upload, and resume history.

Screenshot:

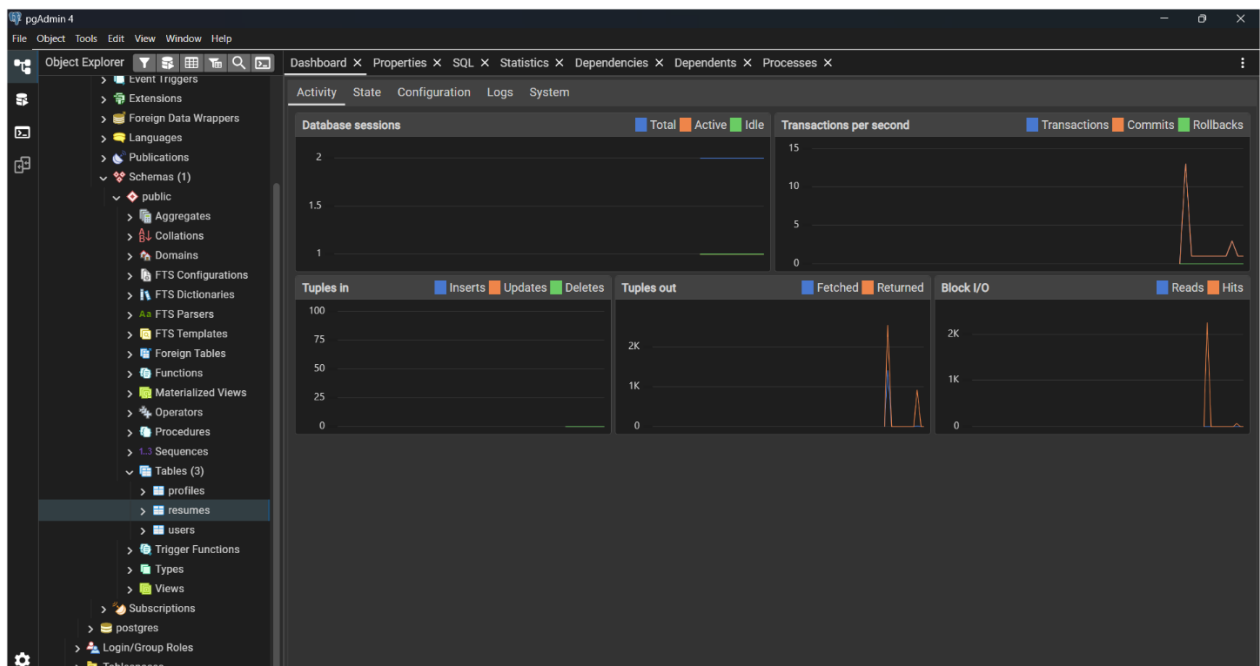


5.10 PostgreSQL Database

Description:

The PostgreSQL database stores user registration information and resume analysis records. SQLAlchemy ORM is used to manage database operations efficiently.

Screenshot:



Final Output

The successful execution of the AI Career Intelligence Platform demonstrates that all the modules developed in Milestone 1 are functioning correctly. The application enables secure user authentication, AI-powered resume analysis, skill gap detection, career recommendation, salary prediction, resume history management, and profile management through an intuitive and responsive user interface.

6. Conclusion

The **AI Career Intelligence Platform** was successfully designed and developed as part of Milestone 1 of the internship project. The application provides a secure and user-friendly platform where users can register, log in, upload resumes, and receive AI-powered resume analysis.

The system successfully extracts important information from resumes, identifies technical skills, calculates a resume score, detects missing skills, recommends suitable career paths, predicts salary ranges, and stores the analysis results in a PostgreSQL database. FastAPI was used to develop efficient REST APIs, while React.js and Tailwind CSS were used to build a modern and responsive user interface.

This project helped strengthen practical knowledge of **full-stack web development, REST API development, database management using PostgreSQL, authentication using JWT, and AI-based resume analysis**. The successful completion of Milestone 1 provides a strong foundation for implementing additional intelligent features in future milestones.

7. Acknowledgment

I would like to express my sincere gratitude to **Infosys Springboard** for providing me with the opportunity to participate in this internship program and work on the **AI Career Intelligence Platform** project.

I am thankful to my mentors and trainers for their continuous guidance, valuable suggestions, and encouragement throughout the development of this project. Their support helped me strengthen my practical knowledge of **Full-Stack Web Development, FastAPI, React.js, PostgreSQL, and AI-based Resume Analysis**.